Paritosh Chauhan
Email-id :chauhan.paritosh01@gmail.com
Mobile No.: 8791741601
Linked In. : @paritosh
Github : @paritoshchauhan01

Profile

CSE B.Tech fourth year student with curiosity, dedication, sound academic background, and a penchant for self-study. Brings a systematic approach to work, a passion for solving problems, and a disciplined attitude that drives him to finish what he starts.

ACADEMIC DETAILS

Examination	Institute	Year	CGPA/%
Graduation	Graphic Era Hill University	2027	8.18

PROJECTS

1. AI-Powered AWS Cost Optimizer

Role: Independently designed and developed the full-stack application — backend API, AWS integration, AI layer, and frontend dashboard.

- Built a full-stack web application that connects to AWS accounts, scans EC2, EBS, RDS, and S3 infrastructure, and fetches real monthly spend via AWS Cost Explorer.
- Developed a multi-turn AI Chat Assistant enabling users to query AWS spending in natural language, improving accessibility for non-technical stakeholders.
- Containerized the full application using Docker and Docker Compose; provisioned cloud infrastructure with Terraform for repeatable, scalable deployments.

Tech Used: Python, FastAPI, React 19, Vite, Recharts, Boto3, AWS Cost Explorer,Groq, Docker, Terraform

2. LMS (Learning Management System)

Role: Independently designed and developed the complete full-stack Learning Management System, including backend APIs, authentication, database architecture, and responsive frontend.

- Developed a role-based LMS with Admin, Teacher, and Student workflows using Google OAuth, JWT, and secure authentication.
- Built REST APIs for courses, assignments, and study materials using Node.js, Express, PostgreSQL, and Sequelize, with Socket.io for real-time updates.
- Deployed the application on Vercel and Render with Neon PostgreSQL, and implemented email-based teacher invitations.

Tech Used: React, Node.js, Express.js, PostgreSQL, Socket.io, JWT, Google OAuth

3. Emotion-Aware Resource Management for OS

Role: Built the entire multiprocessing pipeline, OS listener, real-time socket-mode handler, and did system benchmarking.

- Built an emotion-detecting OS using camera & AI that adapts CPU and memory in real-time.
- Developed a 32-bit kernel with UART serial I/O and dynamic schedulers (SJF, Round Robin, Lottery).
- Created a Python emotion pipeline that switches between Calm / Relaxed / Stressed modes for better performance.

Tech Used: C, NASM, Python, OpenCV, DeepFace, QEMU, UART Serial, JSON Sockets, Tkinter, Linux/Ubuntu

4. Optimal Route Analyser

Role: Integrated the route-optimization algorithm with mapping components and built the full visual map interface.

- Finds the fastest routes between multiple destinations using smart optimization logic.
- Displays fully visible and accurate route paths on a map for easy travel planning.
- Supports multi-stop trip scheduling with data-driven, easy-to-understand route outputs.

Tech Used: Python, NetworkX, Flask, Algorithms, OpenStreetMap

SKILLS

Language: C ,C++,Python
Frameworks & Libraries : React.js , Flask
Databases : MongoDB , MySQL
Tools & Platform : Git , Github , AWS , REST APIs

ACHIEVEMENTS

- AWS Knowledge: Cloud Essentials
- AWS ML Engineer Associate Curriculum Conclusion
- AWS Cloud Quest: Cloud Practitioner